

Asymptotic Notations

O - big oh - upper bound
 Ω - big omega - lower bound
 Θ - theta - average bound

this is the most important.
 but we mostly use big oh notation.

Big Oh

The function $f(n) = O(g(n))$ iff there exist positive constants C and n_0 such that $f(n) \leq Cg(n)$ for all $n \geq n_0$

Example:

Say we have $f(n) = 2n + 3$ and we want to represent it in big oh notation. we need to decide the C and the $g(n)$.

$$2n + 3 \leq ??$$

the best way to choose this is to look at $f(n)$ and then just make it all a multiple

$$2n + 3 \leq 2n + 3n$$

$$2n + 3 \leq 5n \quad \therefore \text{true for all } n \geq 1 \quad \checkmark$$

$\downarrow \quad \downarrow$
 $C \quad g$

We can also choose other values.

$$\begin{aligned}
 2n + 3 &\leq 2n^2 + 3n^2 \\
 2n + 3 &\leq 5n^2 \xrightarrow{C} g \\
 \therefore &\text{also true for } n \geq 1
 \end{aligned}$$

but we can't choose certain functions, $g(n)$.
 And why is that?

b/c we have our inequality as $f(n) \leq Cg(n)$, we want things greater, so $O(n)$, $O(n \log n)$, $O(n!)$ are all correct.

BUT: we choose $O(n)$ b/c it's the closest and that's what we usually do.

$$\underbrace{1 \leq \log n}_{\text{lower bound}} \leq \underbrace{n}_{\text{avg bound}} \leq \underbrace{n \log n \leq n^2 \leq n^3 \leq 2^n \leq n!}_{\text{upper bound}}$$

Big Omega

↳ The function $f(n) = \Omega(g(n))$ iff for some positive constants C and n_0 such that $f(n) \geq Cg(n)$ for all $n \geq n_0$.

This is quite literally the opposite of big oh notation, we see that the inequality is literally flipped. And what this means is, we just want the lower bounds.

$\Omega(n)$
 $\Omega(\log n)$
 $\Omega(1)$

will all be reasonable answers but we want $\Omega(n)$ b/c it's the closest

Big Theta

↳ The function $f(n) = \Theta(g(n))$ iff there exists positive constants C_1, C_2 , and n_0 such that $C_1g(n) \leq f(n) \leq C_2g(n)$.

Example:

Suppose $f(n) = 2n+3$, we know from our two previous examples that lower bound is $1 \times n$ and upper we can say $5 \times n$

$$\begin{array}{ccc}
 1 \times n & \leq 2n+3 & \leq 5 \times n \\
 \downarrow & & \downarrow \\
 g(n) & & g(n)
 \end{array}
 \rightarrow \Theta(n)$$

WE CANNOT SAY ANYTHING ELSE B/C

- if we say our lower bound is anything else other than $1, n \log n$, or n it's wrong. Same goes for upper bound. So the only right answer is n b/c it can be used on both sides, thus it's average.

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①

$f(n) = 2n^2 + 3n + 4$ in big oh notation

$$f(n) \leq cg(n)$$

$$2n^2 + 3n + 4 \leq 2n^2 + 3n^2 + 4n^2$$

$$2n^2 + 3n + 4 \leq 9n^2 \quad \text{for } n \geq 1$$

$$\rightarrow O(n^2)$$

can say anything greater than n^2 too

②

$f(n) = 2n^2 + 3n + 4$ in big omega

$$f(n) \geq cg(n)$$

$$2n^2 + 3n + 4 \geq 1 \times n^2$$

$$2n^2 + 3n + 4 \geq n^2 \quad \text{for } n \geq 1$$

$$\rightarrow \Omega(n^2)$$

can say anything less than n^2 too

③

now for big theta

$$cg(n) \leq f(n) \leq c_2g(n)$$

$$n^2 \leq 2n^2 + 3n + 4 \leq 9n^2$$

$$\rightarrow \Theta(n^2)$$

MUST be this only

NOTE: if we ever encounter $n^2 \log n$, then we need to remember that it goes between n^2 and n^3

⑤

$f(n) = n!$, find everything

upper: $f(n) \leq cg(n)$

$$n! \leq n \times n \times n \times \dots \times n$$

$$n! \leq n^n$$

$$O(n^n)$$

lower: $f(n) \geq cg(n)$

$$n! \geq 1$$

nothing else greater than that, so

$$\Omega(1)$$

now b/c these are different,
we can't find a Θ . So
if we can't, we just
show upper/lower as we did

- big oh is upper. $f(n) \leq cg(n)$
- big omega is lower. $f(n) \geq cg(n)$
- big theta is avg. $cg(n) \leq f(n) \leq cg(n)$

CANNOT MESS
THESE INEQUALITY
SIGNS UP!!

⑤ $f(n) = \log n!$ gives the
same result, we can't
find Θ .

NOTES

- we want the nearest figure, we will always choose what's closest to us. That's the perfect choice.
- Theta means exact.
- Omega and big oh means we're not sure about the exact one.

It's super important to know the inequalities
 $1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^2 \log n < n^3 < 2^n < n^n < n!$

↳ by using this and the problem we have, we can find the reasonable answer. Say we have n^2 as upper. n^2 and to the right is a reasonable answer. If n^2 is also lower n^2 and to the left is reasonable. And n^2 is average.

$O(n^2)$ - upper

$\Omega(n^2)$ - lower

$\Theta(n^2)$ - avg